PROJECT DESIGN PHASE-II
TECHNOLOGY STACK (ARCHITECTURE & STACK)

Team ID: SWTID-2026-1398	**Date:** 30 September 2026
Project Name: Import Data Using Transform Maps	**Maximum Marks:** 4 Marks

Technical Architecture:

The architecture models the ServiceNow automated import pipeline. System Administrators upload employee spreadsheet data (containing Employee ID, Name, Email, Department, and Location) via the ServiceNow Web UI. The data is parsed and ingested into an Import Set Staging Table. Transform Map scripts run transform and coalescing logic using Employee ID as the unique identifier. Matching existing records are updated while new employee entries are inserted into the target Employee Master Table (sys_user), with all logs captured in Import Set Logs.

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1	User Interface	ServiceNow System Administration Portal / Import Set Module for file upload and status monitoring.	ServiceNow Service Portal, HTML5, CSS3, JavaScript
2	Application Logic-1	Import Set Engine for parsing incoming XLSX/CSV employee spreadsheets into staging columns.	ServiceNow Data Sources Engine (JavaScript / GlideRecord)
3	Application Logic-2	Transform Map Engine executing field mapping, coalesce matching on Employee ID, and conditionally triggering Insert or Update actions.	ServiceNow Transform Map Scripts (GlideRecord API, JavaScript)

4	Application Logic-3	Validation & Error Handling logic ensuring non-null Employee ID and valid email formatting before commit.	ServiceNow On-Before / On-After Transform Scripts
5	Database	Temporary Staging Table (`u_imp_employee`) holding raw imported spreadsheet rows.	ServiceNow Import Set Table (Relational DB)
6	Cloud Database	Target Employee Master Table (`sys_user` / Custom Employee Table) storing active employee profiles.	ServiceNow Cloud Platform Database (MariaDB/MySQL backend)
7	File Storage	Storage repository for uploaded raw Excel/CSV employee dataset files.	ServiceNow Sys Attachment Table / Instance Storage
8	External API-1	REST API Integration endpoint allowing automated external batch spreadsheet uploads.	ServiceNow Table API / Import Set API (REST / JSON)
9	External API-2	Notification service triggering email alerts upon successful or failed import batch completion.	ServiceNow Email Notification Engine (SMTP / REST)
10	Machine Learning Model	N/A - Not required for rule-based ETL data transformation.	N/A
11	Infrastructure (Server / Cloud)	Cloud Server Configuration: Enterprise ServiceNow Cloud Instance SaaS deployment.	ServiceNow Now Platform Cloud (Multi-Instance Architecture)

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	Standard web client libraries utilized in ServiceNow UI and Node.js REST API clients.	JavaScript (ES6), HTML5, CSS3, REST Web Services
2	Security Implementations	Role-Based Access Control	ServiceNow Access Control

			(RBAC via `admin` & `import_admin` roles), TLS 1.3 encryption in transit, AES-256 at rest, and OAuth 2.0 authentication for REST APIs.	Lists (ACLs), HTTPS/TLS, OAuth 2.0
	3	Scalable Architecture	ServiceNow high-availability multi-instance architecture supporting async multi-threaded batch transform maps for large bulk datasets.	ServiceNow Import Set Engine & Async GlideRecord Transformer
	4	Availability	High availability enterprise cloud infrastructure backed by redundant data centers and automatic failover.	ServiceNow High Availability (HA) Dual Data Center Architecture
	5	Performance	High-throughput bulk ingestion supporting thousands of records per minute with indexed coalesce fields (Employee ID) to optimize search and update speeds.	Indexed Coalesce Fields, Database Query Optimization

References:

- https://docs.servicenow.com/bundle/utah-platform-administration/page/administer/import-sets/concept/c_ImportDataUsingTransformMaps.html
- https://developer.servicenow.com/dev.do#!/learn/learning-plans/tokyo/servicenow_application_developer/app_store_learnv2_importingdata_tokyo_transform_maps
- https://c4model.com/